

New Pricing Models, Same Old Philips Curves?

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Presented by Mahdi Ebsim

Introduction: Two Models of Sticky Prices

- ▶ **Time-Dependent (TD)**: Probability of price adjustment is exogenously given Ex. Calvo (1983)

→ Easy to implement, gives the familiar New Keynesian Philips Curve

$$\pi_t = \kappa \widehat{mc}_t + \beta \mathbb{E}_t[\pi_{t+1}]$$

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This paper:

- ▶ How close does the PC of TD models get to SD models? Exactly? Numerically?
- ▶ Why is Calvo model a good approximate in the literature?

Results

Define: Generalized Philips Curve (GPC)

- ▶ Mapping from $\widehat{mc} = \{\widehat{mc}_t\}_{t \geq 0}$, real marginal costs shocks, to inflation $\pi = \{\pi_t\}_{t \geq 0}$

$$\pi = K\widehat{mc} \quad \text{or equivalently} \quad \pi_t = \sum_{s \geq 0} K_{t,s} \widehat{mc}_s$$

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- (1) **Menu Cost** GPC = mixture of two **TD** GPCs

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Use to show the following:

- (1) **Menu Cost** GPC = mixture of two **TD** GPCs
- (2) **Menu Cost** GPC $\approx$ Single **Calvo** GPC with properly chosen κ

Outline

State-Dependent Models

Time Dependent Models

GPC for TD

Exact Equivalence between SD and TD

Numerical Equivalence between SD and Calvo

State-Dependent Model: Environment

- ▶ Discrete time, discounted by $\beta \in (0, 1)$, continuum of firms selling at log-price $p_{i,t}$
- ▶ Choose $p_{i,t}$ to match **optimal price** $p_{i,t}^* + \log MC_t$
- ▶ Idiosyncratic shock: $p_{i,t}^* = p_{i,t-1}^* + \varepsilon_{i,t}$ with $\varepsilon_{i,t} \stackrel{\text{iid}}{\sim} F$ with symmetric, single-peaked, mean-zero
- ▶ Firms face **quadratic loss** function: $(p_{i,t} - p_{i,t}^* - \log MC_t)^2$ ▶ NK Model
- ▶ Face **menu costs**, $\xi_{i,t} \in \{0, \bar{\xi}\}$ if $p_{i,t} \neq p_{i,t-1}$, with prob. $\mathbb{P}(\xi_{it} = 0) = \lambda$, iid across i and t
- ▶ Assume **perfect-foresight** of $\log MC_t$ and start from stationary steady state (WLOG $\log MC_0 = 0$)
- ▶ Describe firm's state with **price-gap** $x_{i,t} = p_{i,t} - p_{i,t}^*$

State-Dependent Model: Firm's Problem

$$\min_{\{x_{i,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{1}{2} (x_{i,t} - \log MC_t)^2 + \xi_{i,t} \mathbb{1} \{x_{i,t} \neq x_{i,t-1} - \varepsilon_{i,t}\} \right]$$

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Price-gap follows an (S, s) policy summarized by $(\underline{x}_t, \bar{x}_t, x_t^*)$ where

$$x_{i,t} = \begin{cases} x_{i,t-1} - \varepsilon_{i,t}, & \text{w/ prob. } 1 - \lambda \text{ if } x_{i,t-1} - \varepsilon_{i,t} \in [\underline{x}_t, \bar{x}_t] \\ x_t^*, & \text{otherwise} \end{cases}$$

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- ▶ $(\underline{x}_t, \bar{x}_t)$: inside (S, s) bands price gap is not reset
- ▶ x_t^* : Optimal reset gap
- ▶ Policy $(\underline{x}_t, \bar{x}_t, x_t^*)$ is identical across firms
- ▶ In the steady state: $(\underline{x}, \bar{x}, x^*) = (-\bar{x}, \bar{x}, 0)$.

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Useful definitions:

- ▶ Aggregate Price: $\log P_t = \int x_{i,t} di$
- ▶ Inflation: $\pi_t = \log P_t - \log P_{t-1}$
- ▶ Stationary distribution $g(x)$ of price gaps before adjustment
- ▶ Probability of price adjustment: $\text{freq} = (1 - \lambda) \left(1 - \int_{\underline{x}}^{\bar{x}} g(x) dx \right) + \lambda$

State-Dependent Model

$$V_t(x) = \frac{1}{2}(x - \log MC_t)^2 + \underbrace{\beta(1 - \lambda)\mathbb{E}_\varepsilon \left[\min \left\{ V_{t+1}(x + \varepsilon), \bar{\xi} + \min_{x_{t+1}^*} V_{t+1}(x_{t+1}^*) \right\} \right]}_{\text{No free adjustment}} \\ + \underbrace{\beta\lambda \min_{x_{t+1}^*} V_{t+1}(x_{t+1}^*)}_{\text{Free adjustment}}$$

State-Dependent Model

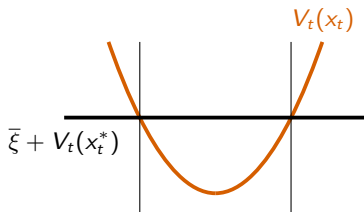
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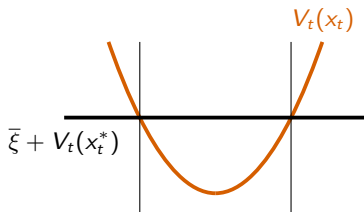
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Criterion

- $V_t(\underline{x}_t) = V_t(\bar{x}_t) = \bar{\xi} + V_t(x_t^*)$
- $V_t'(x_t^*) = 0$

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Time Dependent Models

GPC for TD

Exact Equivalence between SD and TD

Numerical Equivalence between SD and Calvo

Time-Dependent Model: Environment and Policy

- ▶ $\Phi_s \in [0, 1]$ fraction of age- s unadjusted with $\Phi_0 = 1$ and weakly decreasing in s
- ▶ Adjustment Hazard: $\lambda_s \equiv \frac{\Phi_{s-1} - \Phi_s}{\Phi_{s-1}}$
- ▶ Calvo: $\Phi_s = (1 - \lambda)^s \rightarrow \lambda_s = \lambda$ for all s
- ▶ Price gap of firm unadjusted for s -periods: $x_{i,t} - \sum_{r=1}^s \varepsilon_{i,t+r}$

$$\min_{x_{i,t}} \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \Phi_s \left[\frac{1}{2} \left(x_{i,t} - \sum_{r=1}^s \varepsilon_{i,t+r} - \log MC_{t+s} \right)^2 \right]$$

- ▶ The optimal policy is very simple in TD

$$x_{i,t} = \begin{cases} x_t^*, & \text{with probability } \lambda_s, \text{ where } s \text{ is age of current price} \\ x_{i,t-1} - \varepsilon_{i,t}, & \text{otherwise} \end{cases}$$

- ▶ Cohort of firms that change prices together choose the same gap
- ▶ Let a_s be the fraction of firms with s -age prices, $a_s = \frac{\Phi_s}{\sum_{r=0}^{\infty} \Phi_r}$

Layout

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Generalized Phillips Curve: TD

- Solved for a mapping from $\{\log MC_t\}$ to the price-level $\{P_t\}$: $P_t = \mathcal{P}_t(\{\log MC_t\})$

Generalized Phillips Curve: TD

- ▶ Solved for a mapping from $\{\log MC_t\}$ to the price-level $\{P_t\}$: $P_t = \mathcal{P}_t(\{\log MC_t\})$
- ▶ Get log-linearization of P_t as a function of $\{\log MC_t\}$

$$\hat{P}_t = \sum_{s=0}^{\infty} \frac{\partial \log P_t}{\partial \log MC_s} \widehat{MC}_s$$

- ▶ Stack this log-linearization:

$$\hat{P} = \Psi \widehat{MC}$$

where $\Psi_{t,s} = \frac{\partial \log P_t}{\partial \log MC_s}$ and is the **Pass-through** matrix

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- ▶ Convert mapping from $\{\widehat{MC}_t\}$ to the $\{\widehat{P}_t\}$ into mapping from $\{\widehat{mc}_t\}$ shocks to $\{\pi_t\}$

From Pass-through to Phillips Curve

- ▶ $\widehat{mc}_t = \widehat{MC}_t - \widehat{P}_t$

- ▶ $\pi_t = \widehat{P}_t - \widehat{P}_{t-1}$

- ▶ Let $\mathbf{L}$ be a lag matrix with entries of 1 on first diagonal below main diagonal

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Pass-through for TD

Adjusting firm chooses:

$$x_t^* = \frac{\sum_{s=0}^{\infty} \beta^s \Phi_s \log MC_{t+s}}{\sum_{s=0}^{\infty} \beta^s \Phi_s}$$

Price Level:

$$\hat{p}_t = \sum_{s=0}^t a_s x_{t-s}^*$$

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$\Rightarrow$

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- ▶ Price level **weighted average** of past price-gaps
- ▶ Survival function $\{\Phi_t\}_{t \geq 0}$ characterizes $\frac{\partial \hat{P}_t}{\partial MC_s}$, entries of the **pass-through** ▶ [Example](#)

Pass-through for TD

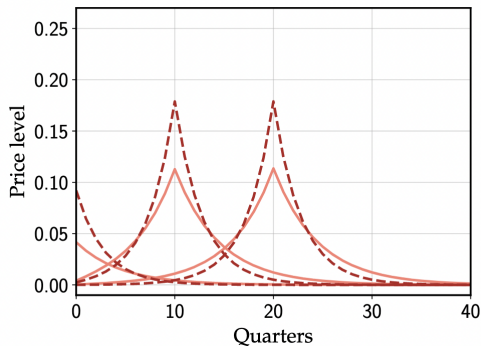
- ▶ Let Ψ^Φ be the pass-through associated with survival function $\{\Phi_s\}$

$$\Psi^\Phi = \frac{1}{\left(\sum_{s \geq 0} \Phi_s\right) \left(\sum_{s \geq 0} \beta^s \Phi_s\right)} \times \begin{bmatrix} \Phi_0 & 0 & 0 & \dots \\ \Phi_1 & \Phi_0 & 0 & \dots \\ \Phi_2 & \Phi_1 & \Phi_0 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \times \begin{bmatrix} \Phi_0 & \beta \Phi_1 & \beta^2 \Phi_2 & \dots \\ 0 & \Phi_0 & \beta \Phi_1 & \dots \\ 0 & 0 & \Phi_0 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

- ▶ **First Matrix** Response to changes in reset prices
- ▶ **Second Matrix** Response of reset prices to future $\log MC_t$ shocks

Pass-through for Calvo

Calvo: Adj Freq 0.2 and 0.3

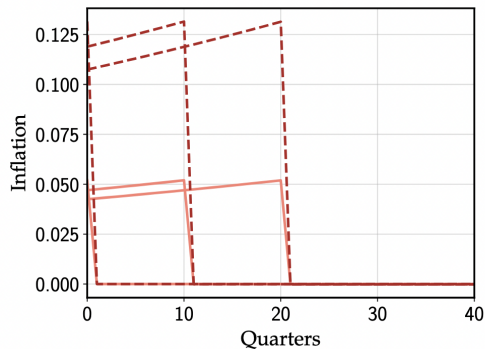


- ▶ Shocks identified by peak
- ▶ Plots the columns of the Ψ for Calvo
- ▶ Greater price flexibility $\rightarrow$ steeper peaks
- ▶ Can trace out survival function of TD model from IRFs

▷ Increasing Hazard Pass-through

Generalized Phillips Curve: Calvo

Calvo: Adj Freq 0.2 and 0.3



$$K = (I - L)(I - \Psi)^{-1}\Psi$$

$$\kappa = \frac{\lambda(1 - \beta(1 - \lambda))}{1 - \lambda}$$

$$K = \begin{bmatrix} \kappa & \beta\kappa & \beta^2\kappa & \dots \\ 0 & \kappa & \beta\kappa & \dots \\ 0 & 0 & \kappa & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

► Increasing Hazard GPC

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Equivalence: SD = Mixture Two TD Models

- ▶ $\Delta \text{ SD Price-level} = \Delta \text{Extensive} + \Delta \text{Intensive}$

Equivalence: SD = Mixture Two TD Models

- ▶ Δ SD Price-level = Δ Extensive + Δ Intensive
- ▶ Extensive: Adjustments to (S, s) bands $[\underline{x}_t, \bar{x}_t]$
 - Reflects shifts in the fraction of firms resetting prices
 - Match these shifts to Ψ^{Φ^e} , a TD model with survival function Φ^e

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 - Reflects shifts in the fraction of firms resetting prices
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- ▶ **Intensive:** Adjustments to reset price-gap x_t^*
 - Reflects the changes in price-gaps conditional on changing the price gap
 - Match these shifts to Ψ^{Φ^i} , a **TD** model with survival function Φ^i

Equivalence: SD = Mixture Two TD Models

Proposition

*The pass-through matrix, Ψ , of the **canonical menu cost** model is a mixture of two time dependent (TD) pass-through matrices:*

$$\Psi = \alpha \Psi^{\phi^e} + (1 - \alpha) \Psi^{\phi^i}$$

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Proposition

The pass-through matrix, Ψ , of the *canonical menu cost* model is a mixture of two time dependent (TD) pass-through matrices:

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The “virtual” survival functions Φ^e and Φ^i and α are such that

- ▶ IRF to permanent MC shock through (S, s) band: $\alpha \sum_{s=0}^t \Phi_s^e / \sum_{s \geq 0} \Phi_s^e$
- ▶ IRF to permanent MC shock through reset price-gap: $(1 - \alpha) \sum_{s=0}^t \Phi_s^i / \sum_{s \geq 0} \Phi_s^i$

Equivalence Result

- ▶ Consider an experiment along (S, s) band and **reset price-gap** separately:
 - At $t - s$, we *perturb* $(\underline{x}_{t-s}, \bar{x}_{t-s}, x_{t-s}^*)$ for one period
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- ▶ Analogously, consider an experiment in $\{d \log MC_t\}_t$
 - At $t + s$, we *perturb* $d \log MC_{t+s}$
 - Study $d\bar{x}_t$, $d\underline{x}_t$, and dx_t^*

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 - Study $d\bar{x}_t$, $d\underline{x}_t$, and dx_t^*
- ▶ Define $E^t(x) = \mathbb{E}[x_t | x_0 = x]$

- Log-linearization of the price-level as a function of shocks to the margins

$$\hat{P}_t = (1 - \lambda)g(\bar{x}) \sum_{s=0}^t E^s(\bar{x}) \underbrace{(d\bar{x}_{t-s} + d\underline{x}_{t-s})}_{\text{Extensive-margin}} + \text{freq} \sum_{s=0}^t E^{s'}(0) \underbrace{dx_{t-s}^*}_{\text{Intensive-margin}}$$

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- Similarly, perturbations in policies as functions of shocks to marginal cost:

$$d\underline{x}_t = d\bar{x}_t = \frac{\sum_{u=0}^{\infty} \beta^u E^u(\bar{x}) d \log MC_{t+u}}{\sum_{u=0}^{\infty} \beta^u E^u(\bar{x})}$$

$$d\underline{x}_t^* = \frac{\sum_{u=0}^{\infty} \beta^u E^{u'}(0) d \log MC_{t+u}}{\sum_{u=0}^{\infty} \beta^u E^{u'}(0)}$$

- Log-linearization of the price-level as a function of shocks to the margins

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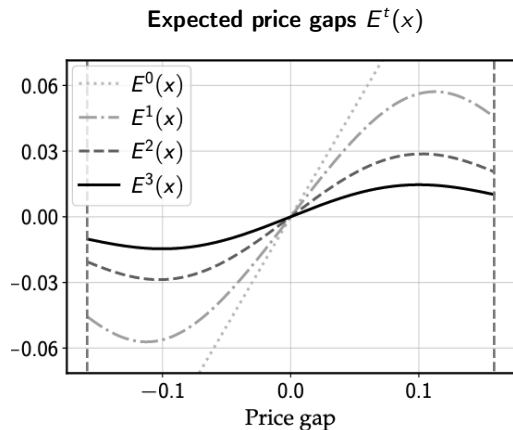
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$$dx_t^* = \frac{\sum_{u=0}^{\infty} \beta^u E^{u'}(0) d \log MC_{t+u}}{\sum_{u=0}^{\infty} \beta^u E^{u'}(0)}$$

$$\hat{P} = \left[2(1 - \lambda)g(\bar{x}) \left(\sum_{u=0}^{\infty} E^u(\bar{x}) \right) \Psi^{\phi_s^e} + \text{freq} \left(\sum_{u=0}^{\infty} E^{u'}(0) \right) \Psi^{\phi_s^j} \right] \widehat{MC}$$

- Define $\phi_s^e = \frac{E^s(\bar{x})}{\bar{x}}$ and $\phi_s^j = E^{s'}(0)$

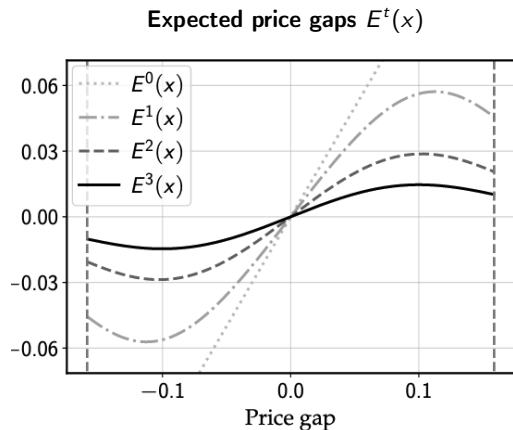
Equivalence Result



Nakamura and Steinsson (2010) calibration

- ▶ How does $d\bar{x}_{t-s}$ effect $\hat{P}_t$?
→ $\hat{P}_t \propto 2(1-\lambda)E^s(\bar{x})g(\bar{x})$
- ▶ How does dx_{t-s}^* effect $\hat{P}_t$?
→ $E^s(dx_{t-s}^*) - E^s(0) = (E^s)'(0)dx_{t-s}^*$
→ $\hat{P}_t \propto \text{freq}(E^s)'(0)$

Equivalence Result



Nakamura and Steinsson (2010) calibration

- ▶ How does $d \log MC_t$ effect $d\bar{x}_{t-s}$?
 $\rightarrow d\bar{x}_{t-s} \propto \beta^s E^s(\bar{x})$
- ▶ How does $d \log MC_t$ effect dx_{t-s}^* ?
 $\rightarrow dx_{t-s}^* \propto \beta^s (E^s)'(0)$
- ▶ Propose the following survival functions

$$\phi_s^e = \frac{E^s(\bar{x})}{\bar{x}} \quad \phi_s^j = E^{s'}(0)$$

Layout

State-Dependent Models

Time Dependent Models

GPC for TD

Exact Equivalence between SD and TD

Numerical Equivalence between SD and Calvo

Estimating Best Calvo Approximation

- ▶ Literature has found that Calvo pricing models approximate many menu cost models well
- ▶ Let K be GPC for a menu cost of interest
- ▶ Let K^{Calvo} be the GPC for a Calvo model with slope κ

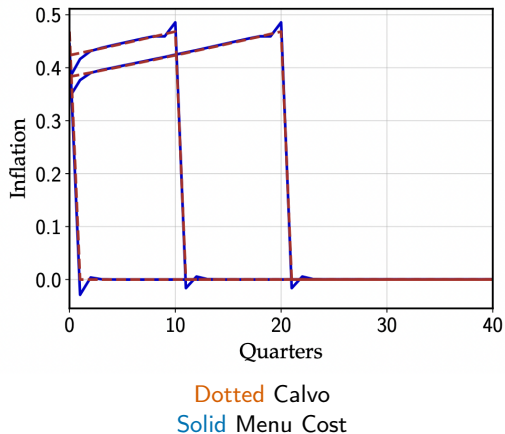
- ▶ Define distance

$$\text{dist}(\mathbf{A}, \mathbf{A}') = \frac{\|\mathbf{A} - \mathbf{A}'\|}{\|\mathbf{A}\|}$$

- ▶ Find κ to minimize $\text{dist}(K^{\text{Calvo}}(\kappa), K)$

Estimating Best Calvo Approximation: Example

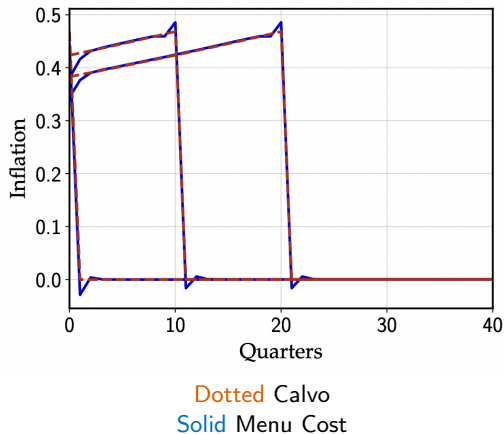
Nakamura and Steinsson (2010)



- ▶ Matched Calvo slope $\kappa = 0.468$
- ▶ IRF decaying at β , backward from the incidence of shocks

Estimating Best Calvo Approximation: Example

Nakamura and Steinsson (2010)



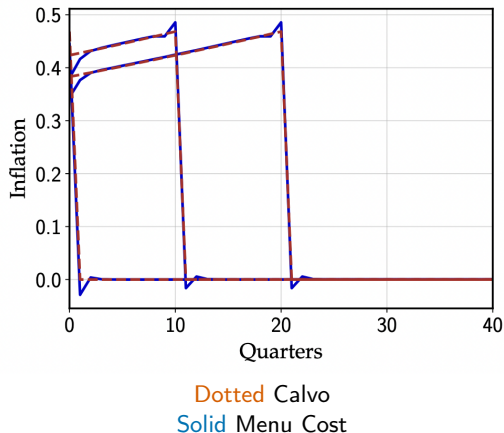
- ▶ Matched Calvo slope $\kappa = 0.468$
- ▶ IRF decaying at β , backward from the incidence of shocks
- ▶ Compare to Calvo:

$$\kappa = \frac{\text{freq}(1 - \beta(1 - \text{freq}))}{1 - \text{freq}} = 0.08$$

with $\beta = 0.99$ and $\text{freq} = 0.239$

Estimating Best Calvo Approximation: Example

Nakamura and Steinsson (2010)



- ▶ Matched Calvo slope $\kappa = 0.468$
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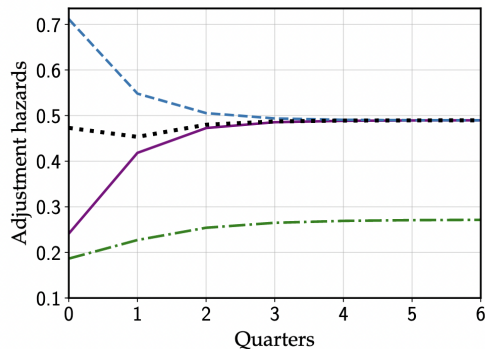
$$\kappa = \frac{\text{freq}(1 - \beta(1 - \text{freq}))}{1 - \text{freq}} = 0.08$$

with $\beta = 0.99$ and $\text{freq} = 0.239$

- ▶ Menu cost models imply greater price flexibility
- ▶ Poor fit around incidence of shocks ▶ [Details](#)

Why Can Calvo Approximate Menu Cost Pricing Models Well?

Nakamura and Steinsson (2010)



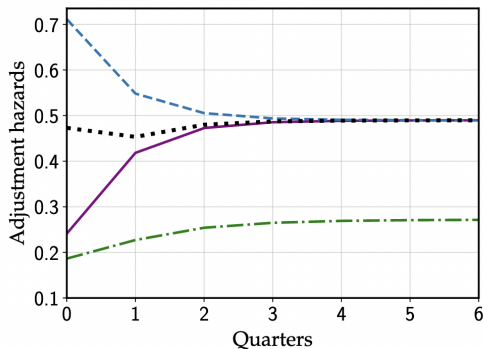
Actual, Virtual, Intensive, and Extensive

▷ Survival Functions

- ▶ Virtual is greater than the actual
- ▶ Actual hazard increasing, virtual doesn't

Why Can Calvo Approximate Menu Cost Pricing Models Well?

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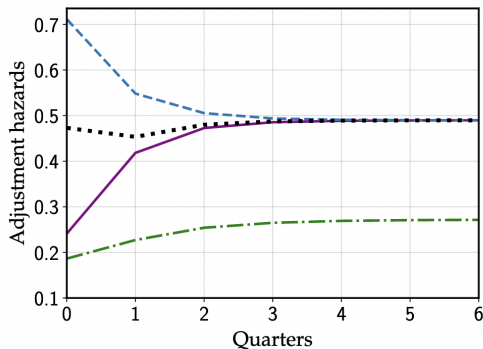
Actual, Virtual, Intensive, and Extensive

▷ Survival Functions

- ▶ Virtual is greater than the actual
- ▶ Actual hazard increasing, virtual doesn't
- ▶ **Result:** $\lambda_{\infty}^v > \lambda_{\infty}^{\text{actual}}$ and $\lambda_{\infty}^e = \lambda_{\infty}^i = \lambda_{\infty}^v$

Why Can Calvo Approximate Menu Cost Pricing Models Well?

Nakamura and Steinsson (2010)



Actual, Virtual, Intensive, and Extensive

▷ Survival Functions

- ▶ Virtual is greater than the actual
- ▶ Actual hazard increasing, virtual doesn't
- ▶ **Result:** $\lambda_{\infty}^v > \lambda_{\infty}^{\text{actual}}$ and $\lambda_{\infty}^e = \lambda_{\infty}^i = \lambda_{\infty}^v$
- ▶ Faster convergence of $\lambda_t^v \Rightarrow$ better approximation ▷ [Example](#)

Conclusion

- ▶ Find an equivalence between GPC of a menu cost model and mixture of two TD models
- ▶ Used the decomposition to explain why menu cost models are well explained by Calvo pricing models
- ▶ Results imply cons of TD models are a concern for SD models

Comments

- ▶ Allow for computation of PC for SD models in linearized environments without relaying on extraneous assumptions
- ▶ Unfortunately, perfect foresight assumption means only low-inflation environments are reflected by the models
- ▶ Micro-founded model not as flexible as it may appear

APPENDIX

One time shock at $t = 0$:

$$\frac{\partial \hat{P}_t}{\partial \widehat{MC}_0} = \frac{\phi_t}{\left(\sum_{s \geq 0} \phi_s\right) \left(\sum_{s \geq 0} \beta^s \phi_s\right)}$$

One time shock at $t = 0$:

$$\frac{\partial \hat{P}_t}{\partial \widehat{MC}_0} = \frac{\Phi_t}{\left(\sum_{s \geq 0} \Phi_s\right) \left(\sum_{s \geq 0} \beta^s \Phi_s\right)}$$

Anticipated shock at $t = 2$, "tent-shapped"

$$\frac{\partial \hat{P}_1}{\partial \widehat{MC}_2} = \frac{\beta \Phi_1 + \Phi_1}{\left(\sum_{s \geq 0} \Phi_s\right) \left(\sum_{s \geq 0} \beta^s \Phi_s\right)}$$

$$\frac{\partial \hat{P}_2}{\partial \widehat{MC}_2} = \frac{\beta \Phi_1^2 + \beta^2 \Phi_2 + 1}{\left(\sum_{s \geq 0} \Phi_s\right) \left(\sum_{s \geq 0} \beta^s \Phi_s\right)}$$

$$\frac{\partial \hat{P}_3}{\partial \widehat{MC}_2} = \frac{\beta \Phi_2 \Phi_1 + \beta^2 \Phi_2 \Phi_3 + \Phi_1}{\left(\sum_{s \geq 0} \Phi_s\right) \left(\sum_{s \geq 0} \beta^s \Phi_s\right)}$$

One time shock at $t = 0$:

$$\frac{\partial \hat{P}_t}{\partial \widehat{MC}_0} = \frac{\Phi_t}{\left(\sum_{s \geq 0} \Phi_s\right) \left(\sum_{s \geq 0} \beta^s \Phi_s\right)}$$

Anticipated shock at $t = 2$, "tent-shapped"

$$\frac{\partial \hat{P}_1}{\partial \widehat{MC}_2} = \frac{\beta \Phi_1 + \Phi_1}{\left(\sum_{s \geq 0} \Phi_s\right) \left(\sum_{s \geq 0} \beta^s \Phi_s\right)}$$

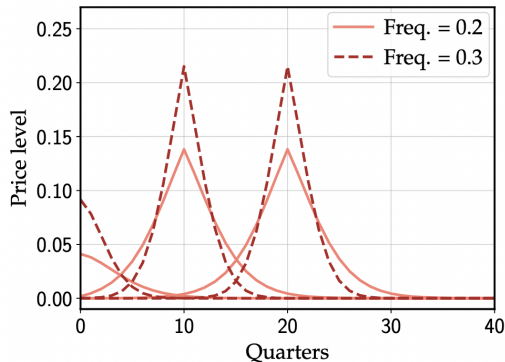
$$\frac{\partial \hat{P}_2}{\partial \widehat{MC}_2} = \frac{\beta \Phi_1^2 + \beta^2 \Phi_2 + 1}{\left(\sum_{s \geq 0} \Phi_s\right) \left(\sum_{s \geq 0} \beta^s \Phi_s\right)}$$

$$\frac{\partial \hat{P}_3}{\partial \widehat{MC}_2} = \frac{\beta \Phi_2 \Phi_1 + \beta^2 \Phi_2 \Phi_3 + \Phi_1}{\left(\sum_{s \geq 0} \Phi_s\right) \left(\sum_{s \geq 0} \beta^s \Phi_s\right)}$$

Permanent shock at $t = 0$

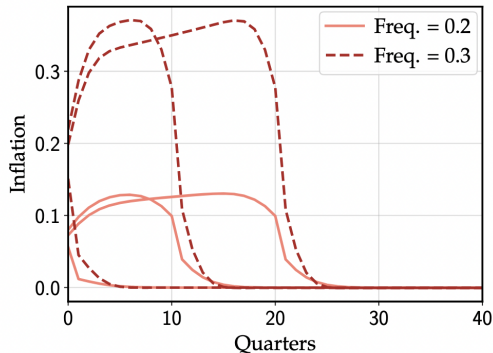
$$\sum_{s=0}^{\infty} \frac{\partial \hat{P}_t}{\partial \widehat{MC}_s} = \frac{\sum_{s=0}^t \Phi_s}{\left(\sum_{s \geq 0} \Phi_s\right)}$$

Increasing Hazard: Total Adj Freq 0.2 and 0.3



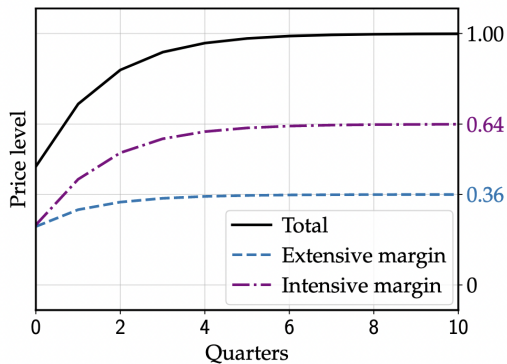
- ▶ Adjustment hazards proportional to $1 - \exp(-0.2(t + 1))$
- ▶ Scaled linearly so that unconditional adjustment frequency equals 0.2 or 0.3

Increasing Hazard: Total Adj Freq 0.2 and 0.3



- ▶ Adjustment hazards proportional to $1 - \exp(-0.2(t + 1))$
- ▶ Scaled linearly so that unconditional adjustment frequency equals 0.2 or 0.3

Nakamura and Steinsson (2010)



Menu cost ($\bar{\xi}$)	0.0513
Prob. of free adj. (λ)	0.179
Shock std. (σ_ε)	0.060
Discount factor (β)	0.99

- ▶ Stack $d\underline{x}_t$ into $d\bar{x}$ and dx_t^* into dx^*
- ▶ Write solutions to $d\bar{x}$ and dx^* in matrix form:

$$d\bar{x} = \frac{1}{\sum_{u=0}^{\infty} \beta^u E^u(\bar{x})} \begin{bmatrix} E^0(\bar{x}) & \beta E^1(\bar{x}) & \beta^2 E^2(\bar{x}) & \dots \\ 0 & E^0(\bar{x}) & \beta E^1(\bar{x}) & \dots \\ 0 & 0 & E^0(\bar{x}) & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \widehat{MC}$$

$$dx^* = \frac{1}{\sum_{u=0}^{\infty} \beta^u E^{u'}(0)} \begin{bmatrix} E^{0'}(0) & \beta E^{1'}(0) & \beta^2 E^{2'}(0) & \dots \\ 0 & E^{0'}(0) & \beta E^{1'}(0) & \dots \\ 0 & 0 & E^{0'}(0) & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \widehat{MC}$$

- Write $\hat{P}_t$ in matrix form as well:

$$\hat{P} = 2(1 - \lambda)g(\bar{x}) \begin{bmatrix} E^0(\bar{x}) & 0 & 0 & \dots \\ E^1(\bar{x}) & E^0(\bar{x}) & 0 & \dots \\ E^2(\bar{x}) & E^1(\bar{x}) & E^0(\bar{x}) & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} d\bar{x} \\ + \text{freq} \begin{bmatrix} E^{0'}(0) & 0 & 0 & \dots \\ E^{1'}(0) & E^{0'}(0) & 0 & \dots \\ E^{2'}(0) & E^{1'}(0) & E^{0'}(0) & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} dx^*$$

- Replace extensive margin matrix entries with $E^s(\bar{x})/\bar{x}$

$$\hat{P} = \frac{2(1-\lambda)g(\bar{x})\bar{x} \left(\sum_{u=0}^{\infty} \frac{E^u(\bar{x})}{\bar{x}} \right)}{\left(\sum_{u=0}^{\infty} \frac{E^u(\bar{x})}{\bar{x}} \right) \left(\sum_{u=0}^{\infty} \beta^u \frac{E^u(\bar{x})}{\bar{x}} \right)} \begin{bmatrix} \frac{E^0(\bar{x})}{\bar{x}} & 0 & 0 & \dots \\ \frac{E^1(\bar{x})}{\bar{x}} & \frac{E^0(\bar{x})}{\bar{x}} & 0 & \dots \\ \frac{E^2(\bar{x})}{\bar{x}} & \frac{E^1(\bar{x})}{\bar{x}} & \frac{E^0(\bar{x})}{\bar{x}} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \begin{bmatrix} \frac{E^0(\bar{x})}{\bar{x}} & \beta \frac{E^1(\bar{x})}{\bar{x}} & \beta^2 \frac{E^2(\bar{x})}{\bar{x}} & \dots \\ 0 & \frac{E^0(\bar{x})}{\bar{x}} & \beta \frac{E^1(\bar{x})}{\bar{x}} & \dots \\ 0 & 0 & \frac{E^0(\bar{x})}{\bar{x}} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \widehat{MC}$$

$$+ \frac{\text{freq} \left(\sum_{u=0}^{\infty} E^{u'}(0) \right)}{\left(\sum_{u=0}^{\infty} E^{u'}(0) \right) \left(\sum_{u=0}^{\infty} \beta^u E^{u'}(0) \right)} \begin{bmatrix} E^{0'}(0) & 0 & 0 & \dots \\ E^{1'}(0) & E^{0'}(0) & 0 & \dots \\ E^{2'}(0) & E^{1'}(0) & E^{0'}(0) & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \begin{bmatrix} E^{0'}(0) & \beta E^{1'}(0) & \beta^2 E^{2'}(0) & \dots \\ 0 & E^{0'}(0) & \beta E^{1'}(0) & \dots \\ 0 & 0 & E^{0'}(0) & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \widehat{MC}$$

- Authors show $E^s(\bar{x})/\bar{x}$ and $E^{s'}(0)$ are between 0 and 1, decreasing in s
- Get familiar upper and lower triangular matrices [▷ For reference](#)

$$\psi^\Phi = \frac{1}{\left(\sum_{s \geq 0} \Phi_s\right) \left(\sum_{s \geq 0} \beta^s \Phi_s\right)} \times \begin{bmatrix} \Phi_0 & 0 & 0 & \dots \\ \Phi_1 & \Phi_0 & 0 & \dots \\ \Phi_2 & \Phi_1 & \Phi_0 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \times \begin{bmatrix} \Phi_0 & \beta \Phi_1 & \beta^2 \Phi_2 & \dots \\ 0 & \Phi_0 & \beta \Phi_1 & \dots \\ 0 & 0 & \Phi_0 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Note

$$E^t(x) = (1 - \lambda) \int_{-\bar{x}}^{\bar{x}} f(x' - x) E^{t-1}(x') dx'$$

Now totally differentiate $V_t(x)$ around the steady state and evaluate at $d \log MC_t$, and for all $s \neq t$ $d \log MC_t = 0$

$$dV_t(x) = -x d \log MC_t$$

$$dV_{t-1}(x) = \beta(1 - \lambda) \int_{-\bar{x}}^{\bar{x}} f(x' - x) dV_t(x)(x') dx'$$

$$dV_{t-s}(x) = \beta(1 - \lambda) \int_{-\bar{x}}^{\bar{x}} f(x' - x) dV_{t-s+1}(x)(x') dx'$$

Comparing to the expectation recursion above: $dV_{t-s}(x) = -\beta^s E^s(x) d \log MC_t$, all evaluated at steady state and price gap x . For any arbitrary shocks $\{d \log MC_s\}_{s \geq 0}$

$$dV_t(x) = - \sum_{u=0}^{\infty} \beta^u E^u(x) d \log MC_s$$

Can show

$$V'_t(x) = x + \beta(1 - \lambda) \int_{-\bar{x}}^{\bar{x}} f(x' - x) V'_{t+1}(x') dx'$$

This is an odd function

$$V'_t(-x) = -x + \beta(1 - \lambda) \int_{-\bar{x}}^{\bar{x}} f(x' - x) V'_{t+1}(-x') dx'$$

Note

$$V'_t(-x) = dV_t(x)|_{d \log MC_s=1}$$

Thus,

$$V'_t(x) = \sum_{u=0}^{\infty} \beta^u E^u(x)$$

► $V_t(\underline{x}_t) = V_t(\bar{x}_t) = \xi + V_t(x_t^*)$

► $V'_t(x_t^*) = 0$

Totally differentiating around steady state

$$dV_t(\bar{x}) + V'_t(\bar{x})d\bar{x}_t = dV_t(0) + V'_t(0)dx_t^*$$

$$d\bar{x}_t = \frac{-dV_t(\bar{x})}{V'_t(\bar{x})}$$

$$dV'_t(0) + V''_t(0)dx_t^* = 0$$

$$dx_t^* = \frac{-dV'_t(0)}{V''_t(0)}$$

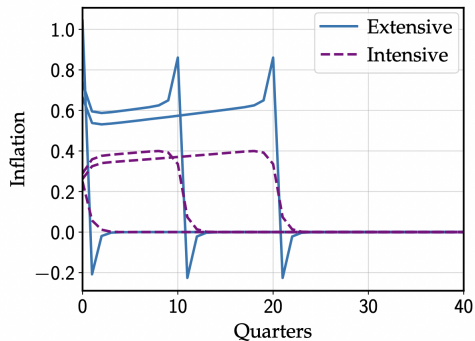
$$\alpha \Psi_{t,0}^{\phi^e} + (1 - \alpha) \Psi_{t,0}^{\phi^i} = \alpha \frac{\phi_0^e \phi_t^e}{\left(\sum_{s \geq 0} \phi_s^e \right) \left(\sum_{s \geq 0} \beta^s \phi_s^e \right)} + (1 - \alpha) \frac{\phi_0^i \phi_t^i}{\left(\sum_{s \geq 0} \phi_s^i \right) \left(\sum_{s \geq 0} \beta^s \phi_s^i \right)}$$

- ▶ Suppose $d \log MC_s = 1$ for all $s \geq 0$. Then $\lim_{t \rightarrow \infty} \hat{P}_t = 1$, $d\bar{x}_t = 1$, and $dx_t^* = 1$

$$1 = 2(1 - \lambda)g(\bar{x}) \left(\sum_{u=0}^{\infty} E^u(\bar{x}) \right) + \text{freq} \left(\sum_{u=0}^{\infty} E^{u'}(0) \right)$$

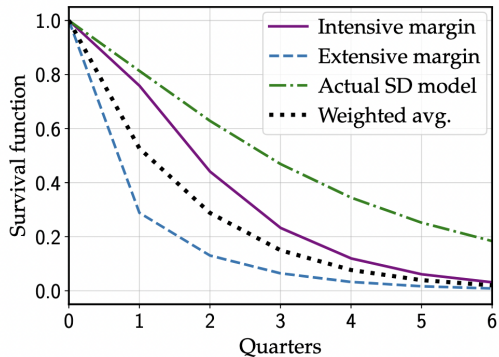
- ▶ let $\alpha = 2(1 - \lambda)g(\bar{x}) \sum_{u=0}^{\infty} E^u(\bar{x})$
- ▶ Permanent shock through [extensive](#) margin: $\alpha \times \sum_{s=0}^{\infty} \Psi_{t,s}^{\Phi^e}$
- ▶ Permanent shock through [intensive](#) margin: $(1 - \alpha) \times \sum_{s=0}^{\infty} \Psi_{t,s}^{\Phi^i}$
- ▶ [▷ One time shock](#)

Nakamura and Steinsson (2010)

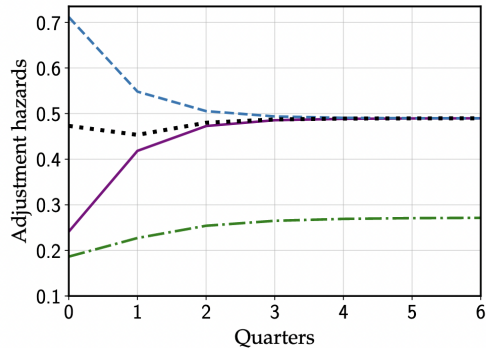


► $\lambda = 0$ is a better fit [▷ Alt. Example](#)

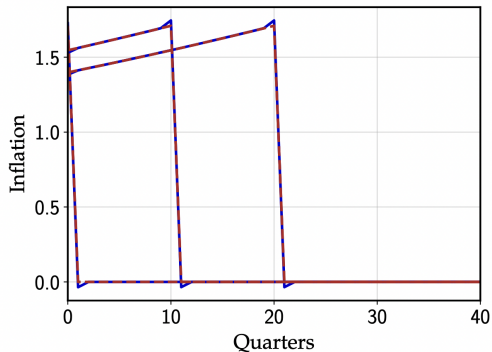
Survival Func. for NS



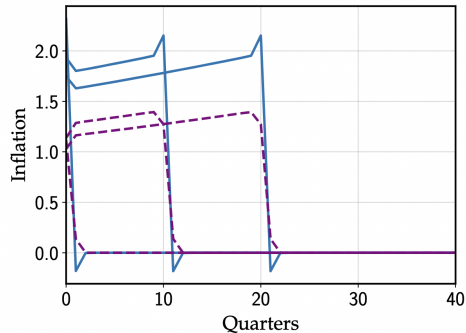
Adj. Hazard for NS



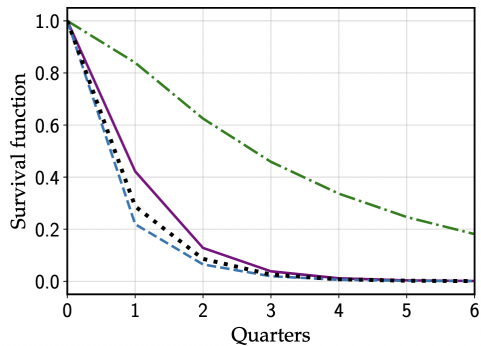
Generalized Philips Curve



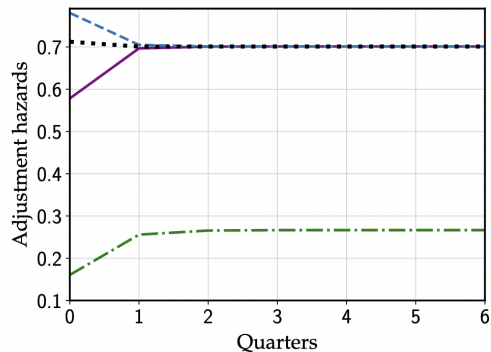
Decomposition of GPC



Survival Func. for GL



Adj. Hazard for GL



General Equilibrium ▶ [Back](#)

- ▶ $\log A_{it}$ is a random walk
- ▶ Household:

$$C_t \equiv \left(\int_0^1 \left(\frac{C_{it}}{A_{it}} \right)^{\frac{\zeta-1}{\zeta}} di \right)$$

- ▶ Firms produce

$$Y_{it} = A_{it} N_{it}$$

- ▶ Face static profits

$$\Pi_{it} = \left(\frac{P_{it}}{P_t} - \frac{W_t}{P_t} \frac{1}{A_{it}} \right) A_{it}^{1-\zeta} \left(\frac{P_{it}}{P_t} \right)^{-\zeta} Y_t$$

- ▶ with optimal price

$$P_{it}^* = \frac{\zeta}{\zeta - 1} \frac{1}{A_{it}}$$

- ▶ Re-write static profits

$$\begin{aligned} \Pi_{it} &= \left(\frac{\zeta}{\zeta - 1} \frac{W_t}{P_t} \right)^{1-\zeta} Y_t \left(\left(\frac{P_{it}}{P_{it}^* W_t} \right)^{1-\zeta} - \frac{\zeta - 1}{\zeta} \left(\frac{P_{it}}{P_{it}^* W_t} \right)^{-\zeta} \right) \\ &= \left(\frac{\zeta}{\zeta - 1} \frac{W_t}{P_t} \right)^{1-\zeta} Y_t \left(e^{(1-\zeta)(x_{it} - \log W_t)} - \frac{\zeta - 1}{\zeta} e^{-\zeta(x_{it} - \log W_t)} \right) \end{aligned}$$